News-Driven Liquidity Dynamics in WTI Crude Oil Futures: A Channel-Decomposition Approach

A Channel-Decomposition Approach

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Motivation

- **News moves markets.** Media sentiment forecasts trading **activity**, not only prices (Tetlock, 2007).
- But most empirical work operates at **daily** resolution, averaging away the intraday window where information actually propagates.
- **WTI crude oil** is an ideal setting: a rich, well-dated event structure (OPEC, geopolitics, the weekly EIA inventory cycle, macro) on top of a deeply liquid market.

> How does news propagate into WTI liquidity at **hourly** resolution? Can we somehow measure the impact? What hits harder the oil liquidity dynamics?

Research questions

RQ1 Lag structure

At what lag does news sentiment exert its strongest effect on WTI trading liquidity?

RQ2 Directional asymmetry

Does the liquidity response differ between **bearish** and **bullish** news?

Liquidity is measured hourly: **log trading volume** (primary), the Amihud illiquidity ratio, and the Parkinson high-low range.

Data

Source	What	Scale
Market	WTI front-month hourly (+ DXY, VIX)	11,232 hours · May 2024 – May 2026
News	GDELT article corpus	76,345 raw → **22,795** unique (19,619 with bodies)
Macro	EIA weekly crude inventories	weekly release cycle

A geopolitical **structural break** (Iran-Israel war onset, early 2026) splits the test window into a **pre-war** and a **war** regime, a natural generalization test.

Liquidity: our ground truth

Every liquidity target is computed **deterministically** from the hourly WTI OHLCV bar (yfinance). These are objective measurements, not model outputs.

Variable	Formula	What it captures
`log_volume`	`ln(volume)`	trading **activity** per hour (primary target)
`amihud`	`abs(log_return) / volume`	**price impact** per unit traded (Amihud 2002)
`price_range`	`ln(high) - ln(low)`	intraday **volatility** (Parkinson range)
`log_return`	`ln(close_t / close_{t-1})`	hourly return (used to calculate `amihud`)

`log_volume`, `amihud`, and `price_range` are the three targets the TFT predicts jointly. `amihud` is undefined for zero-volume hours, which are dropped when it is the target.

General Methodology: a two-phase design

Phase 1: Baseline

Interpretable, classical

FinBERT 3-class sentiment

↓

Lag OLS regressions

Transparent assumptions

→

Phase 2: Extraction

Richer, more expressive

LLM structured extraction
(channels + entities)

↓

Temporal Fusion Transformer

Interpretable deep learning

Neither phase is discarded: **Phase 2 corroborates what Phase 1 discovers through an independent method and enriches the analytical process.**

Phase 1: Method

- **FinBERT** (domain-adapted transformer; Araci, 2019) scores each article: probabilities of **negative**, **neutral**, **positive** tone.
- Articles aligned to the hourly market grid.
- **Lag OLS** at each horizon `k`:

$$\text{log\_volume}_{t+k} = \beta_0 + \beta_1\, P(\text{neg})_t + \beta_2\, P(\text{pos})_t + \varepsilon_t$$

- β_1, β_2: volume response to a maximally confident bearish or bullish article, relative to a neutral one. Corpus: 13,690 articles.

Phase 1: Ordinary Least Squares (OLS)

What it is

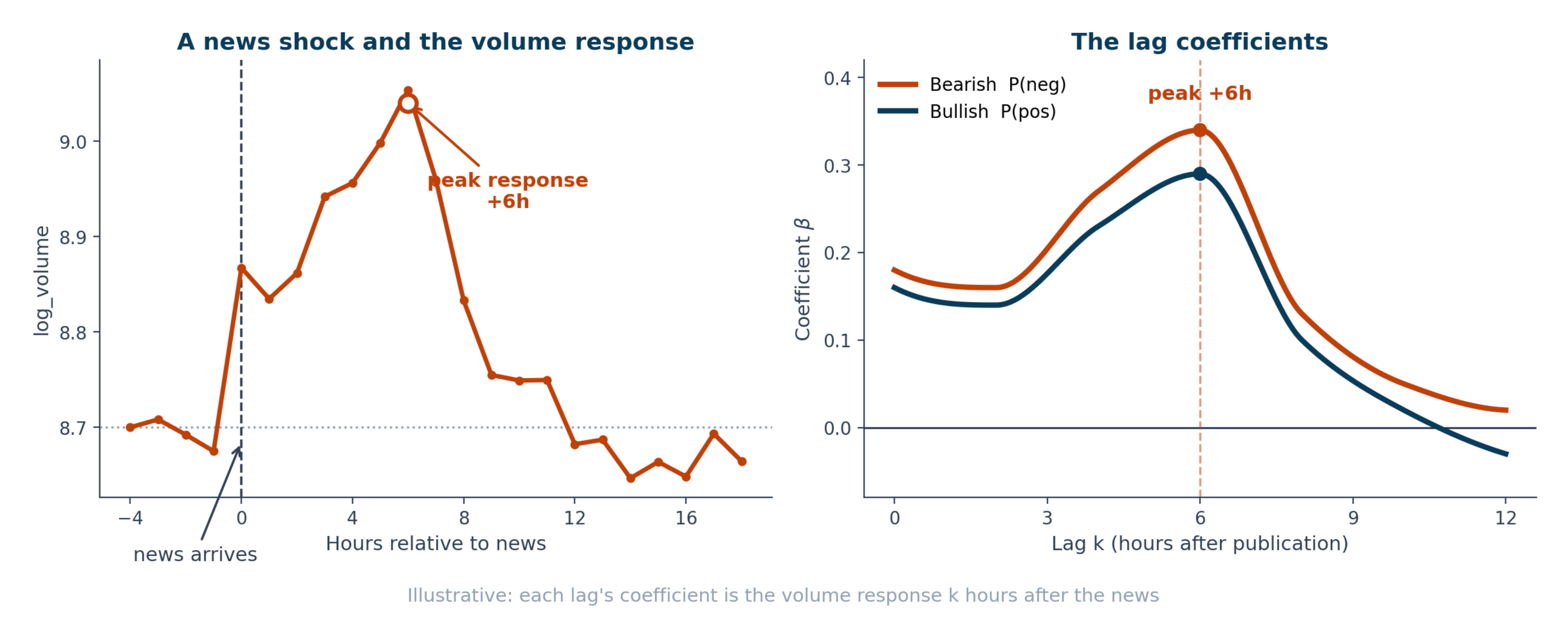
The classical linear regression: fit a linear relationship by minimizing squared error. Here, a **lag regression** of hourly `log_volume` on FinBERT sentiment probabilities, run separately at each horizon `k`.

- **Coefficients** with signs, magnitudes, and **p-values**.
- RQ1 reads off directly: the lag with the largest significant coefficient is the peak.
- RQ2 reads off directly: β_{bearish} vs β_{bullish}.
- Effect sizes in plain units (a bearish article → **+41%** volume at +6h).

Why it was chosen for Phase 1

- **Interpretability first:** the coefficients map one-to-one onto RQ1 (lag) and RQ2 (asymmetry), a clean and defensible first answer.
- **A credible baseline:** transparent, well-understood assumptions; a result that survives a simple model is unlikely to be an artifact.
- **Fits the type of signal:** most hours carry no news; a per-article cross-section handles this event-driven data cleanly.
- **Sets up Phase 2:** its deliberate simplicity (one sentiment axis, one lag at a time) exposes the limits the TFT then addresses.

How the coefficients move



Illustrative dynamics with synthetic data. The real fitted coefficients follow in Phase 1 — Results.

Phase 1: Dataset

Stage	Count
Raw GDELT articles (8 queries)	51,948
Unique (dedup + English)	16,326
Substantive body (`body_valid=1`)	7,756
Title-only fallback	5,934
Modeling dataset	**13,690**
Aligned market hours	11,219

FinBERT sentiment (13,690 articles)

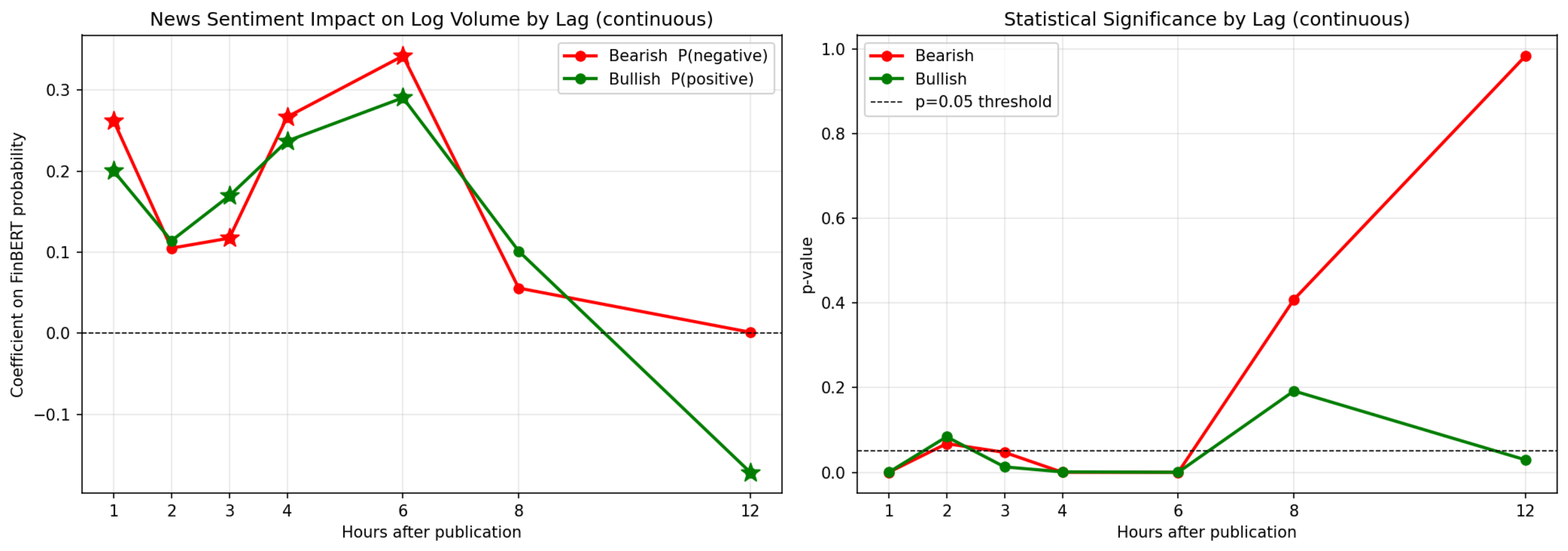
- **44.9%** negative
- **30.4%** positive
- **24.7%** neutral

The bearish skew reflects a headline-tone tendency (see the headline-bias finding), not a population-level claim.

Coverage: March 2024 – February 2026 (pre-war only).

Phase 1: Preliminar results

- **RQ1:** the effect peaks at **+6h** ($\beta_{\text{neg}} = 0.342$, roughly **41% more volume** than a neutral hour), with a secondary trace at +12h. Not contemporaneous.
- **RQ2: bearish > bullish** at the dominant lags (about 18% larger at +6h).



The **p-value** is the probability of seeing a coefficient this large if news truly had no effect on volume. A small value (below 0.05) means the result is unlikely to be a chance artifact, which is why it is the standard gauge of statistical significance. The +6h and +12h peaks clear that threshold; the near-zero lags do not.

Phase 1: Findings and limitations

Findings

- **RQ1:** the news-liquidity response peaks at **+6h**, delayed rather than contemporaneous.
- **RQ2:** a robust **bearish > bullish** asymmetry at the dominant lags.
- **Headline bias:** title vs full-body sentiment disagree on **41.6%** of articles.
- An exploratory **VAR was abandoned**: the hourly news signal is too sparse for a joint time-series model.

Limitations that motivate Phase 2

- FinBERT gives **one sentiment axis** (no magnitude, event type, entities), loses different dimensional information.
- The regex filter is **lexical, not semantic**, high rate of false negatives and positives.
- The signal is **sparse**: about half of hours carry no news.
- The corpus is **pre-war only** (ends Feb 2026).
- The OLS has **no macro controls**, in a complex liquidity dynamic matters

Each limitation maps to a specific Phase 2 change: LLM extraction, a semantic filter, a TFT built for sparse inputs, corpus extension through the war, and DXY/VIX covariates.

Phase 1: Headline bias: Titles vs full bodies

- Same FinBERT model, two inputs: **title only** vs **title + body**.
- They disagree on **41.6%** of articles ($p < 0.001$).
- Systematic, not noise: titles lean **more bullish**; reading the body shifts sentiment bearish (mean signed shift **−0.09**; **57.6%** turn more bearish).
- **Implication:** a title-only pipeline silently inherits this bias, so Phase 2 extracts from **full bodies**.

Divergence (7,755 articles)	Value
Label agreement	4,527 (58.4%)
Label flip	**3,228 (41.6%)**
Flip rate: pos / neu / neg	40.8 / 69.5 / 21.8%
Mean magnitude: flip / same	0.96 / 0.17
Signed shift (title→body)	**−0.09**
More bearish after body	57.6%

Phase 2: The Temporal Fusion Transformer (TFT)

What it is

An attention-based neural network for **multi-horizon** time-series forecasting (Lim et al., 2021), built to be accurate **and** interpretable.

How it works

- Encodes a **48h** window of news + market features, then forecasts **1 / 3 / 6 / 12h** jointly.
- **Variable Selection Networks:** a learned weight per feature (**which** inputs matter).
- **Interpretable attention:** a weight per past hour (**when** the signal sits).
- Outputs **quantiles** (intervals), not just point estimates.

What we expect from it

- A forecaster of hourly liquidity driven by the decomposed news features.
- Two diagnostics read directly off the model: **feature importance** and **temporal attention**.
- To **corroborate RQ1** (the +6h lag) and probe RQ2 through an independent method, triangulating Phase 1 rather than acting as a black box.

Phase 2: Method

- Replace FinBERT with **LLM** structured extraction per article:
  - composite **sentiment** + three orthogonal channels: **supply - demand - risk** (mirroring Kilian's 2009 oil-shock decomposition)
  - magnitude, certainty, event type, and **71 entity flags**
- **Why channels?** A single composite sentiment score disagreed across model families (Haiku vs GPT) on the geopolitical events that move the market. The response: decompose it into three separable channels.
- Model: **Temporal Fusion Transformer** (Lim et al., 2021): 3 targets × 4 horizons, interpretable via variable selection and attention.

Phase 2: Why the LLM filter beats regex

Both filters answer the same question for the same **19,619** articles: **does this carry substantive WTI content?** They agree on **75.2%** (Cohen's $\kappa = 0.47$). The disagreement is where the LLM wins:

Disagreement	Count	What these are
Regex accepts, **LLM rejects**	**3,366**	long, keyword-matched, off-topic (palm oil, canola, equity wraps)
Regex rejects, **LLM accepts**	**1,491**	short but on-topic briefs (e.g. a 320-char OPEC+ note)

- Regex is **blind to topic**: it passes long articles that merely match a keyword (false positives).
- Its 400-character cutoff is **blunt**: it drops short but relevant briefs (false negatives).
- The LLM **reads the body**, so it is correct on both axes. A manual audit confirms its call in the large majority.

Not an oracle: a few residual LLM errors remain (e.g. an EV-tariff article marked usable). Still, it is more accurate than regex on both of the regex's failure modes, so the LLM `usable` flag becomes the canonical Phase 2 filter.

Inter-model calibration: why decompose

Same 30 stratified articles scored independently by two LLMs from different families (**Haiku** and a **GPT reference**). Cross-family disagreement flags genuine ambiguity.

The problem (composite only)

A single sentiment number conflated **event valence** with **price direction**, so the two models read **opposite signs** on the high-magnitude geopolitical events that move the market.

Metric	Composite	+ Channels
Sentiment correlation	0.39	**0.88**
Sign disagreement	31%	**7%**
Supply channel	n/a	0.94
Demand channel	n/a	0.96
Risk channel	n/a	0.82

Decomposing into supply / demand / risk **disciplined the composite** (0.39 → 0.88) and yielded channels two independent models agree on.

Dataset: Phase 1 → Phase 2

	Phase 1	Phase 2
GDELT queries	8	**13**
Raw articles	51,948	**76,345**
Unique articles	16,326	**22,795**
Modeling corpus	13,690	**10,514** (`usable_strict`)
Coverage	to Feb 2026 (pre-war)	to **May 2026** (incl. war)
News features	FinBERT 3-class	**LLM** sentiment + 3 channels + 71 entities
Macro covariates	none	**DXY, VIX**

Phase 2 is **larger and temporally complete** (it now covers the war regime), yet its modeling corpus is **smaller**: a stricter LLM filter and no title-only fallback trade quantity for signal quality.

TFT v2: baseline vs the reported model

Design axis	v2.0 (baseline)	v2.2 exp2 (reported)
Targets	`log_volume`	`log_volume` · `amihud` · `price_range`
Horizons	1h	1 / 3 / 6 / 12h
Categoricals	integer-encoded	proper embeddings
Entity flags	none	**71**

Both train on the same Phase 2 corpus (`usable_strict`, 11,232-hour grid). All reported Phase 2 results use **v2.2 exp2**; v2.0 is the simplest ablation variant. The ablation showed that proper categorical encoding is what unlocks the channels' predictive role.

From headline to features

One of FinBERT's limitation was that news had a lot of features that change how impactful news can be, and condensing that feature set into a single number lost key information.

GDELT article · thehindubusinessline.com · 2 Mar 2026

"Crude oil could top $100 as Strait of Hormuz closure halts flows"

"...Iran warned shipping away from the strait and insurers withdrew coverage, effectively halting tanker movements."

→

LLM structured judgment

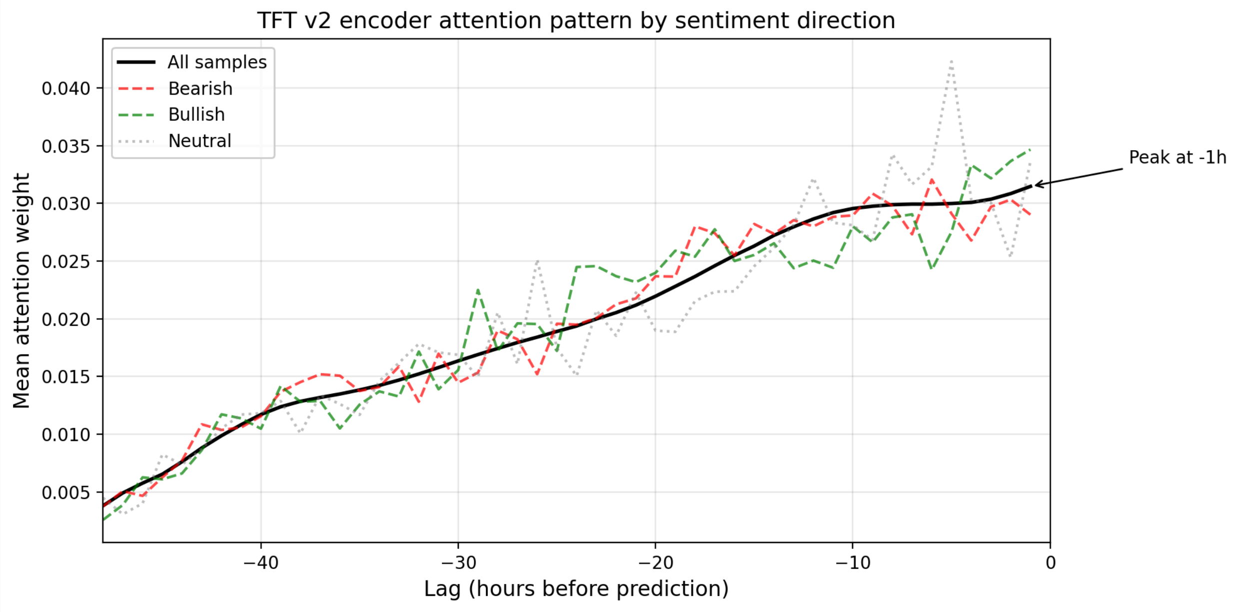

```
{
  "usable": true,
  "sentiment_score": 0.95,
  "supply_impact": -1.0,
  "demand_impact": 0.0,
  "risk_premium": 1.0,
  "magnitude": 1.0,
  "certainty": 0.9,
  "event_type": ["geopolitical", "supply"],
  "time_horizon": "immediate",
  "entities": ["United States", "Israel", "Iran", "Wood Mackenzie", "Strait of Hormuz"]
}
```


A supply threat reads as **bullish for price** (`sentiment_score +0.95`), while the channels record **why**: `supply_impact -1.0`, `risk_premium +1.0`. A single tone score could not separate these.

Phase 2: Results

After a series of experiments varying between dataset features and tuning model params, the **TFT version 2.2 - experiment 2** extends:

- It beats a persistence baseline substantially: **log_volume 46–71%** MAE reduction, amihud 43–45%.
- **Attention:** bearish-sentiment attention peaks at **−6h**, independently corroborating the Phase 1 +6h lag.



Phase 2: Top features (Variable Selection Network)

#	Feature	Weight	Type
1	VIX	0.188	macro
2	supply_impact	0.121	channel
3	ent_oman	0.113	entity
4	demand_impact	0.055	channel
5	is_wednesday	0.022	calendar
6	ent_japan	0.017	entity
7	ent_eu	0.016	entity
8	ent_iran	0.015	entity
9	ent_china	0.014	entity
10	ent_algeria	0.014	entity

The reading

- **VIX** (market-wide volatility) leads.
- Both news **channels** land in the top four (supply, demand).
- **6 of the top 10 are entity flags**, geopolitical actors around the Strait of Hormuz.
- The composite `sentiment_score` does **not** make the top 10.

> **Risk and salience**, not price direction, drive the model.

Feature-**type** dominance is robust; exact per-feature order shifts across retrains.

RQ2, more precisely

- The TFT shows **no** bearish-vs-bullish gap in predicted volume (**0 of 16** tests significant).
- Not a failed replication: the two phases estimate **different quantities**.
  - OLS: a **marginal coefficient** (sensitivity, holding other terms fixed).
  - TFT: a **group-mean** difference (unconditional).
- **Tone vs price:** FinBERT (tone) and Haiku (price) flip sign on geopolitical news. A supply threat is bearish in **tone** but bullish for **price**.

> For the TFT it is not a matter of negative versus positive news, but of the **risk premium** the news implies.

Key findings

- **RQ1:** news propagates into WTI liquidity over a **+6 to +12 hour** window, recovered independently by a linear baseline and a deep-learning forecaster.
- **RQ2:** a robust **bearish > bullish** asymmetry in **marginal sensitivity** (Phase 1); at the level of predicted volume the model organizes around **risk and salience over direction** (Phase 2).
- Supporting findings: the **channel decomposition**, the **LLM-as-filter**, and the **41.6% headline bias**.

Limitations

- **Regime extrapolation:** `price_range` fails on the unseen war regime (trained pre-war only). A clean illustration of a model bound by its training window.
- **Single commodity:** confined to WTI; the risk-and-salience reading may be specific to its geopolitical supply structure.
- **Corpus composition:** GDELT skews to retail-facing web and print, not institutional feeds.
- **Validation:** LLM features are checked by cross-model agreement, not human ground truth.

Conclusion

Three contributions:

- **(a)** An empirical lag-and-asymmetry characterization for WTI at **hourly** resolution.
- **(b)** A **channel decomposition** of LLM-extracted news features, validated by cross-model agreement.
- **(c)** A quantified **title-vs-body headline bias** (41.6%).

Underlying all three: a reusable **two-phase template**, an interpretable baseline plus an expressive model, with neither discarded.

Thank you

Questions?